Vighnesh Das

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Profile

Computer Science and Engineering (Cyber Security) undergraduate at VIT with an **8.91 CGPA**, focused on cybersecurity, secure backend engineering, applied AI, and cloud-native systems. Hands-on experience spans **Android-to-cloud security engineering, Google Cloud deployment, applied cryptography, computer vision, blockchain-oriented systems, and Linux security**. Contributor to **two patent applications** covering multi-modal synthetic-media detection and blockchain-based cultural-artifact provenance.

Experience

VIT-AP – Summer Internship 2026

Android Security, Full-Stack Engineering & Cloud Deployment

- Engineered and hardened **SourceForm**, an Android-to-cloud secure document-intake platform using Kotlin/Android, Node.js/Express, Firebase and Google Cloud services.
- Implemented **hybrid authentication**: Firebase Authentication with Google, Facebook and GitHub OAuth, centralized backend token verification, JWT-based authorization, inactivity timeout and protected API routes.
- Built defense-in-depth controls including **Fernet-compatible metadata encryption, SHA-256 integrity verification, deny-by-default private storage, short-lived signed URLs, audit logging, magic-number file validation, size limits, rate limiting and hardened error handling**.
- Deployed the backend on **Google Cloud Run** with Firestore and Firebase Storage/Google Cloud Storage; worked with HTTPS, environment-based configuration, managed secret injection and scoped cloud IAM/service identities.
- Performed negative and end-to-end validation of invalid/expired tokens, cross-user access, tampered files, direct storage access, signed-URL expiry, disguised uploads and session timeout; resolved authentication, storage and deployment-configuration issues.

Patent Applications

A Multi-Modal Image Authenticity Verification System for Distinguishing Real Images from Artificially Generated Images

- Contributed to a multi-stream authenticity architecture combining **pixel-level classifier ensembles, statistical/frequency/noise forensic analysis, semantic consistency reasoning, feature aggregation and gradient-boosted meta-classification**.
- Implemented the research architecture as **MMAIFA** in Python with modular pixel models, forensic streams, semantic reasoning, fusion, calibration, uncertainty/OOD analysis, provenance and explainability; the public repository uses FastAPI and a React forensic dashboard.
- The patent specification reports a CIFAKE experimental result of **99.53% accuracy, 99.11% sensitivity and 99.60% specificity**; the system also supports confidence-based escalation of low-confidence cases for human review.

Blockchain-Based System for Cultural Artifact Provenance Verification and Digital Marketplace Integration

- Contributed to a blockchain-oriented provenance and marketplace system for **tamper-resistant artifact records, authenticity/ownership traceability and digital marketplace integration**.
- The associated **Dynasty** repository contains the web marketplace/application layer, including artifact/artist dashboards, marketplace, cart/checkout, authentication and JavaScript-driven application workflows.

Selected Engineering Projects

SourceForm – Secure Cloud Intake Platform *Kotlin, Node.js, Express, Firebase, GCP*

- Built a secure document-submission pipeline: authentication → validation → integrity hashing → metadata encryption → audit logging → private cloud storage → signed retrieval.

Fernet-REST-Service – Cryptographic Microservice *Node.js, REST, Docker*

- Developed a standalone Dockerized Fernet-compatible service implementing **AES-128-CBC, HMAC-SHA256, PKCS7 padding, timestamp/TTL validation, tamper detection and URL-safe Base64**; exposes encryption, decryption, key-generation and metadata endpoints.

SecureDocs Platform *TypeScript, Express, PostgreSQL, Prisma, Docker, Nginx, Authentik, Redis*

- Built a self-hosted security-first document platform with containerized backend infrastructure, PostgreSQL/Prisma persistence, Nginx reverse proxy and Authentik OIDC identity integration; current public repository records **Phase 2.3 – Identity Integration**.

Simple DiffBot Gazebo / ROS 2 *ROS 2, URDF, Gazebo, RViz2*

- Developed a differential-drive robot simulation package covering URDF modeling, robot-state publishing, Gazebo simulation, RViz2 visualization and ROS 2 launch/build workflows.

Cybersecurity Credentials & Practice

TryHackMe – Pre Security Learning Path *Completed 13 Aug 2026*

- Completed the **Pre Security** learning path (19 h 10 min), covering networking, Linux, web fundamentals and practical security concepts.

PreGrad – Cyber Security *01 Jun 2025 – 01 Sep 2025*

- Credential **CCC1685**; completed a structured cybersecurity learning program.

Practical Security

- Worked with **Kali Linux, Nmap, Wireshark and Metasploit** for reconnaissance, enumeration, vulnerability analysis, exploitation and privilege-escalation exercises in CTF/lab environments.
- Configured **SNORT IDS**, developed detection rules and analyzed alerts for suspicious network activity.

Technical Skills

Languages: Python, C, C++, Java, JavaScript, Kotlin, SQL, Verilog HDL
Cybersecurity: Kali Linux, Nmap, Wireshark, Metasploit, SNORT, OWASP, Network Security, Vulnerability Analysis
Backend/Web: Node.js, Express, REST APIs, React, HTML, CSS
Cloud/DevOps: Google Cloud Run, Firestore, Firebase Storage, Docker, Docker Compose, Nginx, Prisma, PostgreSQL, Redis, Git
AI/ML: PyTorch, Computer Vision, Deepfake Detection, FastAPI, Model Evaluation, Feature Fusion, Uncertainty/OOD Analysis
Other: Blockchain, Smart Contracts, ROS 2, Gazebo, RViz2, MATLAB, RStudio, CIS-ORCAD

Education

Vellore Institute of Technology (VIT), Vellore 2024–2028
B.Tech in Computer Science and Engineering (Cyber Security) **CGPA: 8.91**
Relevant Coursework: Computer Networks, Cyber Security, Machine Learning, Data Structures and Algorithms, Digital Systems Design, Microprocessors and Microcontrollers, Software Engineering

Lakshmi School (ICSE), Madurai
Class XII – 95.8% Class X – 96.7%

Additional Projects on GitHub

- **MoodMusicGeneratorRemastered** – software project for mood-driven music generation.
- **ALU-6-Functions** – Verilog digital-system project implementing a six-operation ALU.
- **mission-control-ros2** – ROS 2 project extending the robotics work beyond the DiffBot simulation.
- **BeastQuest-Patch** – software patching/engineering project.

Achievements

- **Two patent applications** involving AI-based image authenticity verification and blockchain-based cultural-artifact provenance/marketplace systems.
- **Institute-level engineering ideathon winner**, contributing to a technical solution that progressed toward patent filing.